

MID TERM EXAMINATION

Subject: Linked Lists, Stacks, Queues

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 38 | Date: 12-03-2026

Total Questions: 10 | Answer all questions.

Section 1: MCQ (4 Questions - 8 Marks)

Q1. [2 marks] (Easy)

Which of the following is NOT a fundamental operation performed on a linked list?

Q2. [2 marks] (Easy)

A Stack data structure follows which principle?

Q5. [2 marks] (Medium)

In an array-based implementation of a circular queue, if front points to the first element and rear points to the last element, what condition indicates that the queue is empty? Assume MAX_SIZE is the capacity.

Q7. [2 marks] (Medium)

Which of the following is a primary advantage of using a linked list over a dynamic array for data storage?

Section 2: SHORT ANSWER (6 Questions - 30 Marks)

Q3. [5 marks] (Easy)

Explain one common real-world application where a Queue data structure is typically used.

Q4. [5 marks] (Medium)

Describe the step-by-step process to insert a new node at the beginning of a singly linked list. Assume head points to the first node and the new node is newNode.

Q6. [5 marks] (Medium)

Differentiate between a Stack and a Queue data structure based on their primary access mechanisms and typical use cases.

Q8. [5 marks] (Medium)

What is a Doubly Linked List, and state one significant advantage it offers compared to a Singly Linked List?

Q9. [5 marks] (Hard)

In a circular queue implemented using an array, it's often tricky to distinguish between a full queue and an empty queue just by checking front and rear pointers. Explain one common strategy used to correctly identify both full and empty conditions in an array-based circular queue.

Q10. [5 marks] (Hard)

A program needs to manage a list of tasks. New tasks can arrive at any time and should be added to the list. The program frequently needs to perform two operations: (1) quickly find and remove the task with the highest priority, and (2) iterate through all tasks in the order they were initially added. Which data structure (from Linked Lists, Stacks, Queues) would you recommend for this scenario, and why is it problematic for both operations? Suggest a combination or alternative if none are suitable.

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